

2. Implement Kubernetes deployment

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ayushi2323@LAPTOP-6LVIDNNHT:~/kubernetes-app$ kubectl cluster-info
Kubernetes control plane is running at https://127.0.0.1:63219
CoreDNS is running at https://127.0.0.1:63219/api/v1/namespaces/kube-system/
services/kube-dns:dns/proxy

To further debug and diagnose cluster problems, use 'kubectl cluster-info du
mp'.
ayushi2323@LAPTOP-6LVIDNNHT:~/kubernetes-app$ kubectl get nodes
NAME                                STATUS    ROLES    AGE   VERSION
desktop-control-plane               Ready    control-plane   46m   v1.36.1
ayushi2323@LAPTOP-6LVIDNNHT:~/kubernetes-app$ nano deployment.yaml

apiVersion: apps/v1
kind: Deployment
metadata:
  name: myapp
spec:
  replicas: 1
  selector:
    matchLabels:
      app: myapp
  template:
    metadata:
      labels:
        app: myapp
    spec:
      containers:
      - name: myapp
        image: nginx:1.25
        ports:
        - containerPort: 80
```

```
ayushi2323@LAPTOP-6LVIDNNHT:~/kubernetes-app$ kubectl apply -f deployment.yaml
deployment.apps/myapp created
ayushi2323@LAPTOP-6LVIDNNHT:~/kubernetes-app$ kubectl get deployments
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
myapp                0/3     3             0           7s
nginx-deployment    2/2     2             2           19m
ayushi2323@LAPTOP-6LVIDNNHT:~/kubernetes-app$ kubectl get deployments
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
myapp                0/3     3             0           14s
nginx-deployment    2/2     2             2           19m
ayushi2323@LAPTOP-6LVIDNNHT:~/kubernetes-app$ kubectl scale deployment myapp
--replicas=5
deployment.apps/myapp scaled
```

```
ayushi2323@LAPTOP-6LVIDNNHT:~/kubernetes-app$ kubectl get pods
NAME                                READY   STATUS              RESTARTS   A
GE
myapp-7cf674d55c-5qqtp             0/1     ContainerCreating   0           2
6s
myapp-7cf674d55c-chzwm             0/1     ContainerCreating   0           5
s
myapp-7cf674d55c-d2hng             0/1     ContainerCreating   0           5
s
myapp-7cf674d55c-dzbtn             0/1     ContainerCreating   0           2
6s
myapp-7cf674d55c-h9jg9             0/1     ContainerCreating   0           2
6s
nginx-deployment-59c4c87bc6-5cgx5  1/1     Running             0           1
9m
nginx-deployment-59c4c87bc6-gf4wj  1/1     Running             0           1
9m
ayushi2323@LAPTOP-6LVIDNNHT:~/kubernetes-app$ kubectl set image deployment/m
yapp myapp=nginx:1.27
deployment.apps/myapp image updated
```

```

ayushi2323@LAPTOP-6LVIDNHT:~/kubernetes-app$ kubectl rollout status deployment/myapp
Waiting for deployment "myapp" rollout to finish: 3 out of 5 new replicas have been updated...
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```

deployment "myapp" successfully rolled out

```

ayushi2323@LAPTOP-6LVIDNHT:~/kubernetes-app$ kubectl get pods

```

NAME	READY	STATUS	RESTARTS	AGE
myapp-78b68469f9-gk45k	1/1	Running	0	47s
myapp-78b68469f9-h678x	1/1	Running	0	3m5s
myapp-78b68469f9-knf2p	1/1	Running	0	3m5s
myapp-78b68469f9-qx6jq	1/1	Running	0	48s
myapp-78b68469f9-zv4qj	1/1	Running	0	3m5s
nginx-deployment-59c4c87bc6-5cgx5	1/1	Running	0	22m
nginx-deployment-59c4c87bc6-gf4wj	1/1	Running	0	22m

```

ayushi2323@LAPTOP-6LVIDNHT:~/kubernetes-app$ kubectl rollout history deployment/myapp

```

```

deployment.apps/myapp
REVISION  CHANGE-CAUSE
1          <none>
2          <none>

```

```

ayushi2323@LAPTOP-6LVIDNHTT:~/kubernetes-app$ kubectl rollout undo deployment/myapp
Warning: resource deployments/myapp was previously managed with 'kubectl apply'. Rolling back will not update the kubectl.kubernetes.io/last-applied-configuration annotation, which may cause unexpected behavior on future 'kubectl apply' operations. Consider using 'kubectl apply' with your previous configuration file instead.
deployment.apps/myapp rolled back
ayushi2323@LAPTOP-6LVIDNHTT:~/kubernetes-app$ kubectl rollout status deployment/myapp
deployment "myapp" successfully rolled out
ayushi2323@LAPTOP-6LVIDNHTT:~/kubernetes-app$ kubectl describe deployment myapp
Name:                                myapp
Namespace:                           default
CreationTimestamp:                   Sat, 26 Sep 2026 05:15:43 +0000
Labels:                              <none>
Annotations:                         deployment.kubernetes.io/revision: 3
Selector:                            app=myapp
Replicas:                            5 desired | 5 updated | 5 total | 5 available | 0 unavailable
StrategyType:                        RollingUpdate
MinReadySeconds:                     0
RollingUpdateStrategy:               25% max unavailable, 25% max surge
Pod Template:
  Labels:  app=myapp
  Containers:
    myapp:
      Image:          nginx:1.25
      Port:           80/TCP
      Host Port:      0/TCP
      Environment:    <none>
      Mounts:         <none>
  Volumes:           <none>
  Node-Selectors:    <none>
  Tolerations:       <none>
Conditions:
  Type           Status  Reason
  ----           -
  Available      True    MinimumReplicasAvailable
  Progressing    True    NewReplicaSetAvailable
OldReplicaSets:  myapp-78b68469f9 (0/0 replicas created)
NewReplicaSet:   myapp-7cf674d55c (5/5 replicas created)
Events:

```